

Experiment 1: Working with Python Packages – NumPy, SciPy, Scikit-learn, Matplotlib

Exploratory Data Analysis on the Iris and Loan Approval Datasets

1 Aim

The aim of this experiment is to explore core Python data-science libraries – NumPy, Pandas, SciPy, Scikit-learn, and Matplotlib/Seaborn – by applying them to real datasets downloaded from public repositories, and to perform Exploratory Data Analysis (EDA) on these datasets using summary statistics and visual tools such as histograms, box plots, scatter plots, pair plots, and correlation heatmaps.

2 Datasets

Two datasets were used to demonstrate the EDA workflow:

- **Iris Dataset** (`Iris.csv`) – 150 samples of iris flowers with 4 numeric features (sepal length, sepal width, petal length, petal width) and a categorical target, **Species**, with 3 classes. This is a classic multi-class **classification** problem.
- **Loan Approval Dataset** (`loan_approval_dataset.csv`) – loan applicant records with numeric attributes (income, loan amount, CIBIL score, asset values, etc.) and a categorical target, **loan_status** (Approved/Rejected). This is a binary **classification** problem.

3 EDA Implementation

A reusable `perform_eda()` function was written to automate EDA for any tabular dataset. Given a dataframe and an (optional) target column, it:

1. Strips column-name whitespace and drops any identifier column (`id`, `index`, `Unnamed: 0`).
2. Prints the dataset shape, column data types, missing-value counts per column, and summary statistics (`describe()`).
3. Imputes missing numeric values with the column median.
4. Auto-detects the target column (by name keywords such as `species`, `status`, `class`, `amount`, etc., or by low cardinality) and infers the problem type (classification if the target is categorical or has ≤ 20 unique values, else regression).
5. Generates a combined multi-panel figure – per-feature histograms (with KDE), per-feature box plots grouped by target class, and a correlation heatmap – saved as a single `.eps` file.

6. Prints per-feature descriptive statistics: mean, median, standard deviation, min, max, quartiles, skewness, and kurtosis.

A representative snippet of the function is shown below:

```
def perform_eda(df, dataset_name="dataset", output_dir="./images",
               font_path="./times.ttf", target_col=None):
    df = df.rename(columns=lambda c: c.strip())
    # ... drop id column, impute missing values, detect target & problem
    # type ...

    fig = plt.figure(figsize=(6 * n_cols, 15))
    gs = fig.add_gridspec(3, n_cols)

    for i, col in enumerate(cols):
        # Row 1:
        # histograms
        ax = fig.add_subplot(gs[0, i])
        sns.histplot(df[col], kde=True, ax=ax)

    for i, col in enumerate(cols):
        # Row 2: box
        # plots by target
        ax = fig.add_subplot(gs[1, i])
        sns.boxplot(x=target_col, y=col, data=df, ax=ax)

    ax_heat = fig.add_subplot(gs[2, :])
    # Row 3:
    # correlation heatmap
    sns.heatmap(df[numeric_cols].corr(), annot=True, cmap="coolwarm", ax=
ax_heat)

    plt.savefig(f"{output_dir}/{dataset_name}_classification_eda.eps",
               format="eps", dpi=600)
```

Both datasets were passed through this function:

```
df1 = pd.read_csv("Iris.csv")
perform_eda(df1, dataset_name="iris")

df2 = pd.read_csv("loan_approval_dataset.csv")
perform_eda(df2, dataset_name="loan")
```

4 Iris Dataset – EDA

The Iris dataset has no missing values and is perfectly balanced (50 samples per species). Figure 1 shows the automated classification-EDA panel: histograms of the four numeric features, box plots of each feature grouped by species, and the feature correlation heatmap.

Figure 2 extends this with a fuller combined view of the same distributions and relationships across all features, and Figure 3 shows a Seaborn pair plot, which visualizes pairwise scatter relationships between every pair of features, colored by species – making the near-perfect linear separability of *Iris-setosa* from the other two species immediately visible, especially along the petal length and petal width dimensions.

Observations:

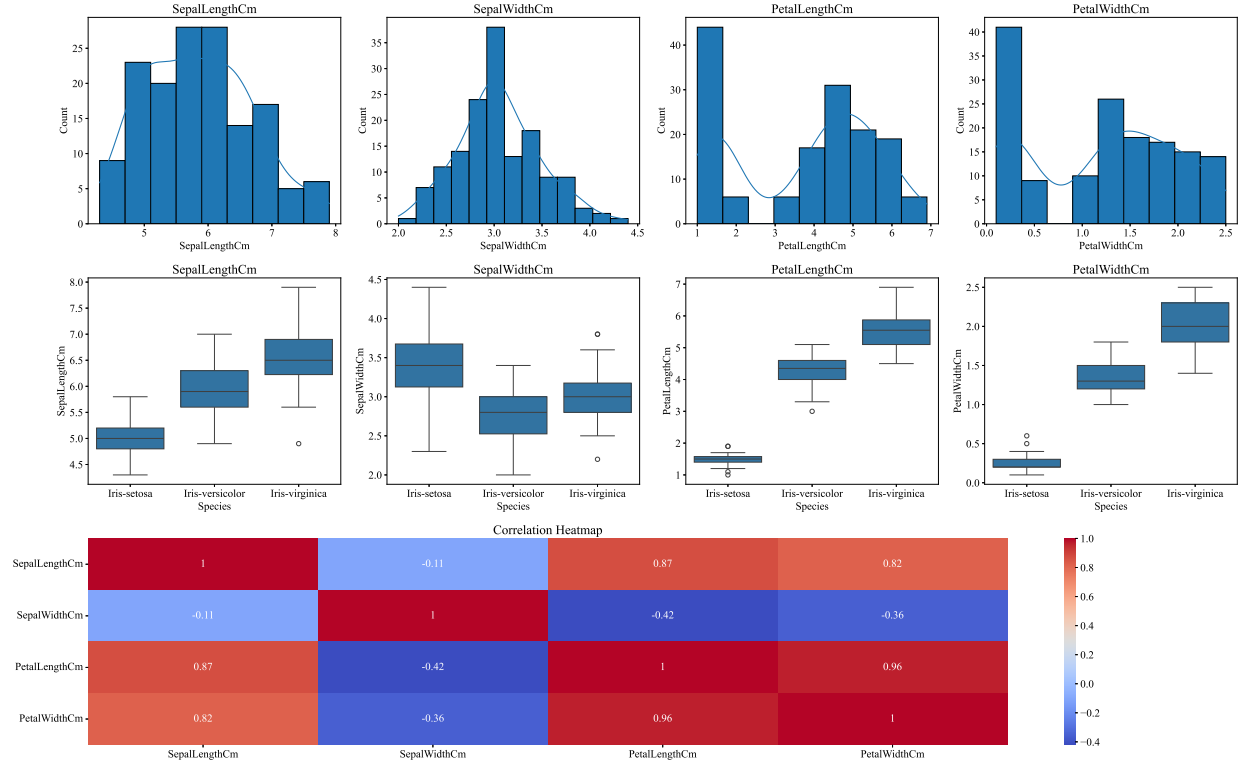


Figure 1: Iris dataset – automated EDA panel: (top row) feature histograms with KDE, (middle row) feature box plots grouped by **Species**, (bottom row) correlation heatmap of the four numeric features.

- Petal length and petal width are strongly correlated with each other and are the most discriminative features for separating the three species.
- Sepal width shows the weakest correlation with the other features and the most overlap across species.
- *Iris-setosa* is linearly separable from *Iris-versicolor* and *Iris-virginica* on nearly every feature, while the latter two show some overlap, particularly in sepal measurements.

5 Loan Approval Dataset – EDA

The loan approval dataset is larger and noisier than Iris, with financial attributes spanning very different scales (income and asset values in the hundreds of thousands to millions, CIBIL score in the hundreds, number of dependents in single digits). Figure 4 shows the automated classification-EDA panel, and Figure 5 shows the expanded combined view.

Observations:

- CIBIL score shows a visibly different distribution between the Approved and Rejected classes, making it one of the strongest predictors of loan status.
- Income, loan amount, and asset-value features are right-skewed, with a long tail of high-value applicants.

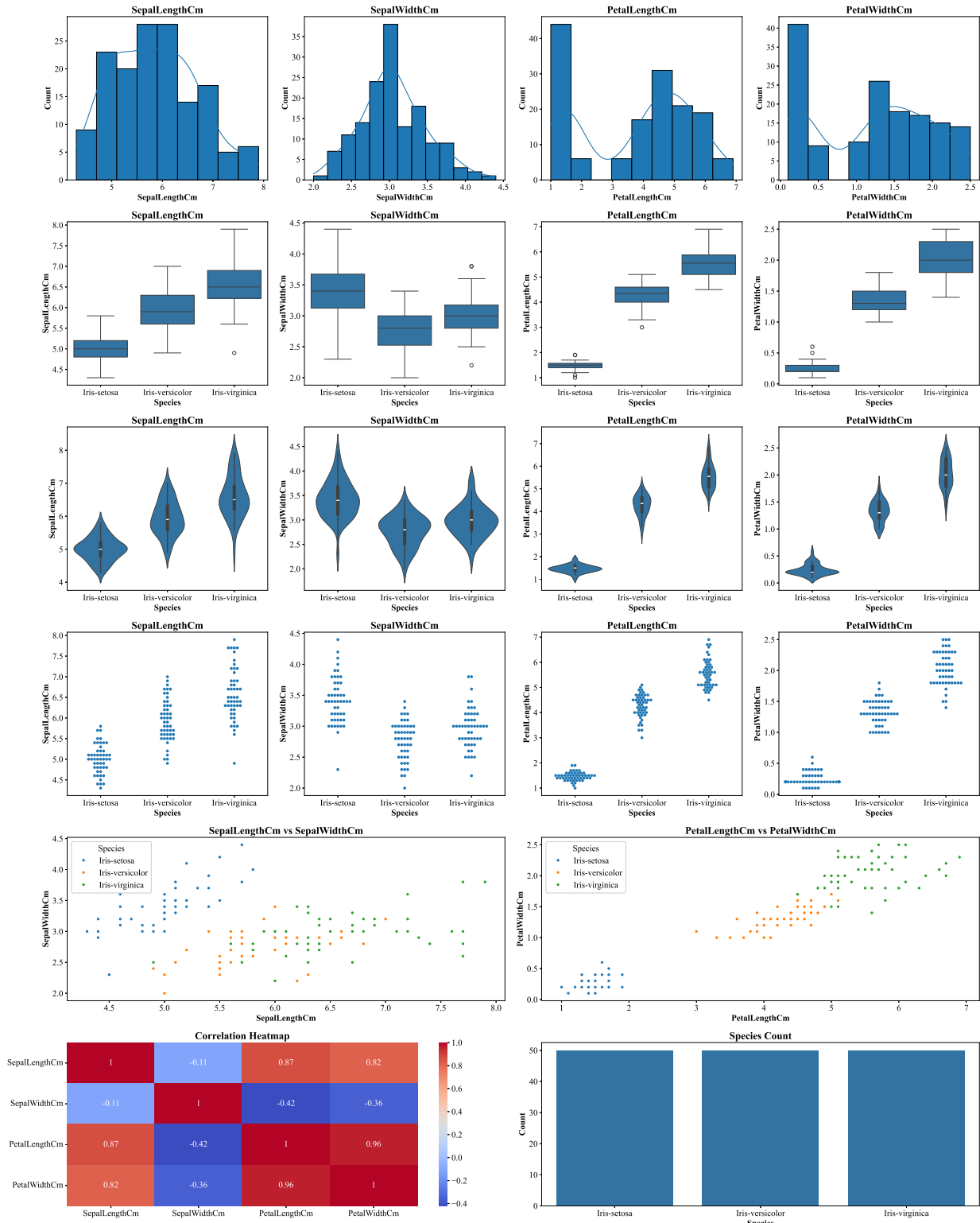


Figure 2: Iris dataset – combined EDA panel with an expanded view of feature distributions, class-wise comparisons, and correlations.

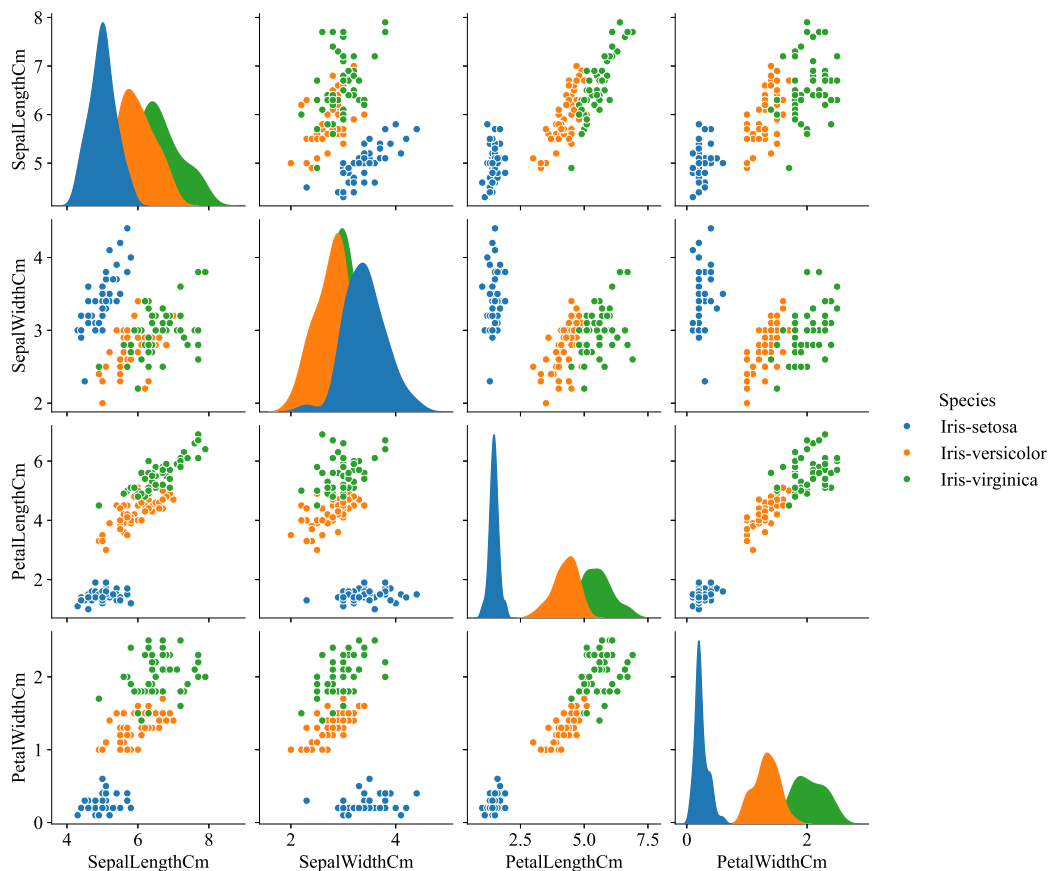


Figure 3: Iris dataset – pair plot of all four numeric features, colored by **Species**, showing pairwise scatter relationships and per-feature distributions along the diagonal.

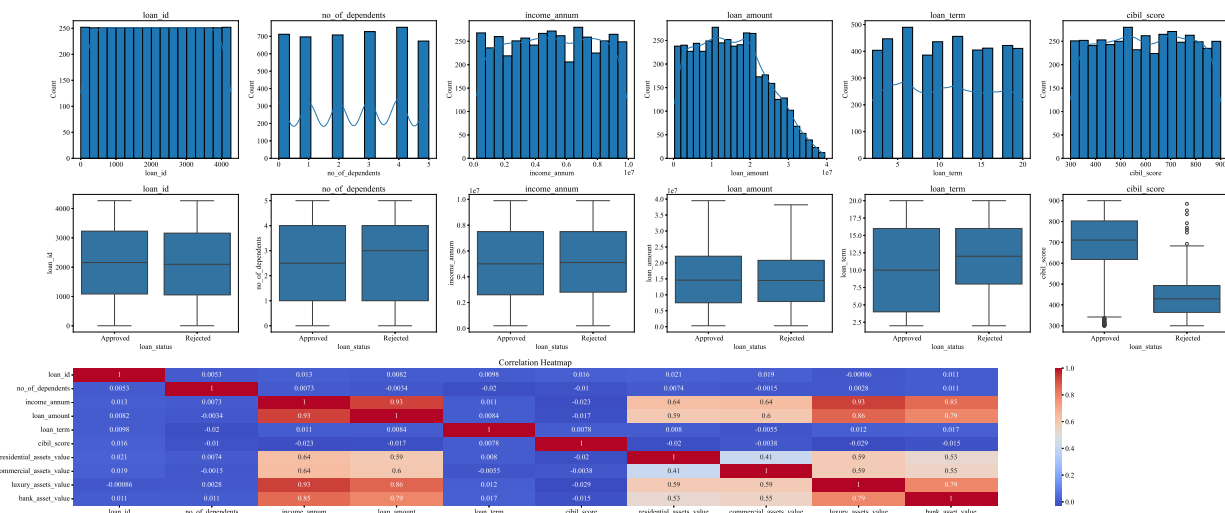


Figure 4: Loan approval dataset – automated EDA panel: (top row) feature histograms with KDE, (middle row) feature box plots grouped by **loan_status**, (bottom row) correlation heatmap of the numeric features.

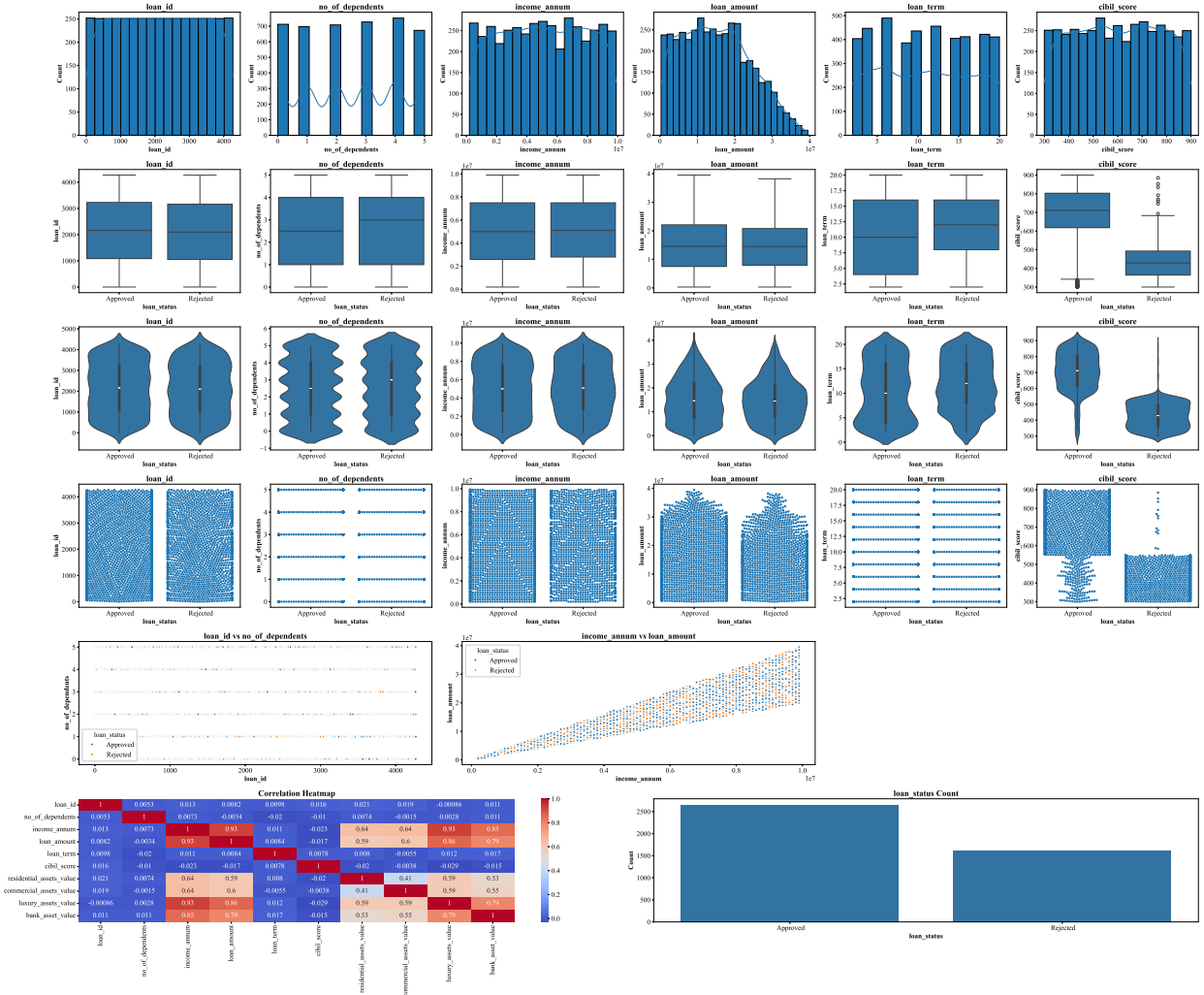


Figure 5: Loan approval dataset – combined EDA panel with an expanded view of feature distributions, class-wise comparisons, and correlations.

- Several asset/financial features are moderately-to-strongly correlated with one another (e.g. income with loan amount, and the various asset value columns with each other), indicating some redundancy that could be addressed with feature selection or dimensionality reduction before modeling.

6 Inference

Both datasets required only light preprocessing (median imputation for missing numeric values; no missing values were actually present in either dataset) before EDA could be performed. The Iris dataset is small, clean, and well-separated by class, making it well-suited to simple classifiers (e.g. Logistic Regression, KNN, Decision Trees). The loan approval dataset is larger, has features on very different scales, and shows skewed distributions and inter-feature correlations, indicating that feature scaling and possibly feature selection would be beneficial before training a classifier. In both cases, EDA identified the target column, the relevant numeric features, and the key relationships

between features and target that guide the subsequent choice of model and preprocessing pipeline.